

Machine Learning Paradigms and Learning Types

Machine learning encompasses several distinct learning paradigms, each suited for different problem types. Supervised learning trains models using labeled data where inputs are paired with correct outputs. The algorithm learns to map inputs to outputs by minimizing prediction errors. Regression predicts continuous values like prices or temperatures, while classification assigns discrete categories like spam versus legitimate emails. Regression examples include predicting house prices or stock values, while classification encompasses disease diagnosis and sentiment analysis. Unsupervised learning discovers hidden patterns in unlabeled data without predefined output categories. Clustering groups similar data points together, revealing natural structure. Dimensionality reduction techniques compress data to essential features, improving visualization and computational efficiency. Anomaly detection identifies unusual or outlier patterns. Semi-supervised learning combines limited labeled data with abundant unlabeled data, improving performance when labeling is expensive. Reinforcement learning trains agents through interaction with environments, receiving rewards for desirable actions and penalties for undesirable ones. The agent learns optimal decision-making policies maximizing cumulative reward. Self-supervised learning creates training labels from data structure itself, enabling learning from massive unlabeled datasets. Transfer learning leverages knowledge from one domain to solve problems in related domains. Active learning strategically selects which data to label next, minimizing labeling costs.